

OPERATING SYSTEMS CONCEPT – LAB

EXP – 05: SCHEDULING ALGORITHMS – FCFS, SJF, RR AND PRIORITY

FCFS ALGORITHM

```
Activities Terminal Apr 20 17:43
rathnavelu06@localhost:~/rathna/scheduling — /usr/bin/vim fcfs.c

#include <stdio.h>
int main(){
    int n;
    printf("Enter number of processes: ");
    scanf("%d", &n);
    int at[n], bt[n], ct[n], tat[n], wt[n];
    for(int i=0; i<n; i++){
        printf("Enter Arrival time and Burst time for P%d: ", i+1);
        scanf("%d %d", &at[i], &bt[i]);
    }
    float total_tat=0, total_wt=0;
    ct[0]=at[0]+bt[0];
    for(int i=0; i<n; i++){
        if(ct[i-1]<at[i]){
            ct[i]=at[i]+bt[i];
        } else{
            ct[i]=ct[i-1]+bt[i];
        }
    }
    for(int i=0; i<n; i++){
        tat[i]=ct[i]-at[i];
        wt[i]=tat[i]-bt[i];
        total_tat+=tat[i];
        total_wt+=wt[i];
    }
    printf("Average Turnaround Time: %.2f", total_tat/n);
    printf("\nAverage Waiting Time: %.2f\n", total_wt/n);
    return 0;
}
```

```
Activities Terminal Apr 20 17:45
rathnavelu06@localhost:~/rathna/scheduling

[rathnavelu06@localhost scheduling]$ vi fcfs.c
[rathnavelu06@localhost scheduling]$ gcc fcfs.c -o fcfs
[rathnavelu06@localhost scheduling]$ ./fcfs
Enter number of processes: 4
Enter Arrival time and Burst time for P1: 0 2
Enter Arrival time and Burst time for P2: 1 2
Enter Arrival time and Burst time for P3: 5 3
Enter Arrival time and Burst time for P4: 6 4
Average Turnaround Time: 3.50
Average Waiting Time: 0.75
[rathnavelu06@localhost scheduling]$
```

SJF ALGORITHM

```
Activities Terminal Apr 20 17:46
rathnavelu06@localhost:~/rathna/scheduling — /usr/bin/vim sjf.c

#include <stdio.h>
int main(){
    int n;
    printf("Enter number of processes: ");
    scanf("%d", &n);
    int at[n],bt[n], ct[n], tat[n], wt[n];
    int done[n];
    for(int i=0; i<n; i++){
        printf("Enter Arrival Time and Burst Time for P%d: ", i+1);
        scanf("%d %d", &at[i], &bt[i]);
        done[i]=0;
    }
    int completed=0;
    int completionTime=0;
    float avgtat=0, avgwt=0;
    while(completed<n){
        int idx=-1;
        int minBT=9999;
        for(int i=0; i<n; i++){
            if(at[i]<=completionTime && done[i]==0){
                if(bt[i]<minBT){
                    minBT=bt[i];
                    idx=i;
                }
            }
        }
        if(idx!=-1){
            if(completionTime<at[idx]){
                completionTime=at[idx];
            }
            completionTime+=bt[idx];
            tat[idx]=completionTime-at[idx];
            wt[idx]=tat[idx]-bt[idx];
            done[idx]=1;
            completed++;
        } else{
            completionTime++;
        }
    }
    for(int i=0; i<n; i++){
        avgtat+=tat[i];
        avgwt+=wt[i];
    }
    printf("\nAverage Turnaround Time: %.2f\n", avgtat/n);
    printf("Average Waiting Time: %.2f\n", avgwt/n);
    return 0;
}
```

```
Activities Terminal Apr 20 17:48
rathnavelu06@localhost:~/rathna/scheduling

[rathnavelu06@localhost scheduling]$ vi sjf.c
[rathnavelu06@localhost scheduling]$ gcc sjf.c -o sjf
[rathnavelu06@localhost scheduling]$ ./sjf
Enter number of processes: 4
Enter Arrival Time and Burst Time for P1: 1 3
Enter Arrival Time and Burst Time for P2: 2 4
Enter Arrival Time and Burst Time for P3: 1 2
Enter Arrival Time and Burst Time for P4: 4 4

Average Turnaround Time: 6.25
Average Waiting Time: 3.00
[rathnavelu06@localhost scheduling]$
```

PRIORITY ALGORITHM

```
Activities Terminal Apr 20 17:50
rathnavelu06@localhost:~/rathna/scheduling — /usr/bin/vim priority.c

#include <stdio.h>
int main(){
    int n, completed=0, time=0;
    printf("Enter number of processes:\n");
    scanf("%d", &n);
    int at[n], bt[n], priority[n], done[n];
    int ct[n], tat[n], wt[n];
    float avgtat=0, avgwt=0;
    for(int i=0; i<n; i++){
        printf("Enter AT, BT and priority for P%d: ", i+1);
        scanf("%d %d %d", &at[i], &bt[i], &priority[i]);
        done[i]=0;
    }
    while(completed<n){
        int idx=-1;
        int highPriority=9999;
        for(int i=0; i<n; i++){
            if(at[i]<=time && done[i]==0){
                if(priority[i]<highPriority){
                    highPriority=priority[i];
                    idx=i;
                } else if(priority[i]==highPriority){
                    if(at[i]<at[idx])
                        idx=i;
                }
            }
        }
        if(idx!=-1){
            time+=bt[idx];
            ct[idx]=time;
            tat[idx]=ct[idx]-at[idx];
            wt[idx]=tat[idx]-bt[idx];
            done[idx]=1;
            completed++;
        } else{
            time++;
        }
    }

    for(int i=0; i<n; i++){
        avgtat+=tat[i];
        avgwt+=wt[i];
    }
    printf("\nAverage Turnaround Time: %.2f\n", avgtat/n);
    printf("Average Waiting Time: %.2f\n", avgwt/n);
    return 0;
}
```

```
Activities Terminal Apr 20 17:51
rathnavelu06@localhost:~/rathna/scheduling

[rathnavelu06@localhost scheduling]$ vi priority.c
[rathnavelu06@localhost scheduling]$ gcc priority.c -o priority
[rathnavelu06@localhost scheduling]$ ./priority
Enter number of processes:
4
Enter AT, BT and priority for P1: 0 4 2
Enter AT, BT and priority for P2: 0 3 1
Enter AT, BT and priority for P3: 0 2 3
Enter AT, BT and priority for P4: 0 1 4

Average Turnaround Time: 7.25
Average Waiting Time: 4.75
[rathnavelu06@localhost scheduling]$
```

ROUND ROBIN ALGORITHM

```
Activities Terminal Apr 20 17:48
rathnavelu06@localhost:~/rathna/scheduling — /usr/bin/vim rr.c

#include <stdio.h>
int main(){
    int n, tq, completed=0, completionTime=0;
    printf("Enter number of processes:\n");
    scanf("%d", &n);
    printf("Enter Time Quantum: \n");
    scanf("%d", &tq);
    int at[n], bt[n], rem[n], tat[n], wt[n];
    float avgtat=0, avgwt=0;
    for(int i=0; i<n; i++){
        printf("Enter AT and BT for P%d: ", i+1);
        scanf("%d %d", &at[i], &bt[i]);
        rem[i]=bt[i];
    }
    while(completed<n){
        int idle=1;
        for(int i=0; i<n; i++){
            if(at[i]<=completionTime && rem[i]>0){
                idle=0;
                if(rem[i]>tq){
                    completionTime+=tq;
                    rem[i]-=tq;
                } else{
                    completionTime+=rem[i];
                    tat[i]=completionTime-at[i];
                    wt[i]=tat[i]-bt[i];
                    rem[i]=0;
                    completed++;
                }
            }
        }
        if(idle){
            completionTime++;
        }
    }
    for(int i=0; i<n; i++){
        avgtat+=tat[i];
        avgwt+=wt[i];
    }
    printf("\nAverage Turnaround Time: %.2f\n", avgtat/n);
    printf("Average Waiting Time: %.2f\n", avgwt/n);
    return 0;
}
```

```
Activities Terminal Apr 20 17:50
rathnavelu06@localhost:~/rathna/scheduling

[rathnavelu06@localhost scheduling]$ vi rr.c
[rathnavelu06@localhost scheduling]$ gcc rr.c -o rr
[rathnavelu06@localhost scheduling]$ ./rr
Enter number of processes:
4
Enter Time Quantum:
2
Enter AT and BT for P1: 0 5
Enter AT and BT for P2: 1 4
Enter AT and BT for P3: 2 2
Enter AT and BT for P4: 4 1

Average Turnaround Time: 7.25
Average Waiting Time: 4.25
[rathnavelu06@localhost scheduling]$
```